JOHN PAUL

Computer Science | Full-Stack & Backend Engineering

Harrisonburg, VA • (804) 424-0730 • john@jgpiv.com • www.linkedin.com/in/johngpaul/ • GitHub • Website

SUMMARY

Computer Science student and co-founder/full-stack engineer building a legal-tech platform with TypeScript, Next.js, Express, PostgreSQL, Supabase, and Docker. Experience spans backend APIs, authorization, financial data integrity, database design, and systems programming in C.

EXPERIENCE

Daubert — Co-Founder & Full-Stack Engineer | Legal-Tech Startup 2026 - Present

- **Architected and developed a pilot-stage legal-tech platform** using TypeScript, Next.js, React, Express, PostgreSQL, Supabase, and Docker across authentication, engagement management, compensation records, and responsive expert/attorney workflows.
- **Designed engagement-scoped authorization** for lead experts, collaborators, and attorneys, separating organization membership from matter-level access and enforcing sensitive record isolation at the server/query layer.
- **Built an auditable compensation ledger** using frozen engagement rate schedules, integer-cent financial calculations, immutable historical records, and centralized retainer and authorized-budget calculations.
- **Developed a CSV ingestion and review workflow** for Harvest/QuickBooks-style exports with source detection, column mapping, validation, corrective review, server-authoritative rate resolution, and transactional row publishing.
- **Built an end-to-end authorization readiness suite** covering unauthenticated access, cross-engagement isolation, staged-record visibility, CSV permissions, matter-scoped access, and platform-administrator boundaries.

EDUCATION

B.S. in Computer Science • James Madison University • Expected Dec. 2026

***Relevant Coursework**: Software Engineering, Data Structures & Algorithms, Computer Systems, Database Systems, Web Development, Data Analytics*

***Honors:** President's List, Dean's List*

A.A. in Social Sciences • J. Sargeant Reynolds Community College • May 2023

SELECTED SOFTWARE PROJECTS

Virtual Memory & CPU Simulation (Mini-ELF) — C | Computer Systems

- **Implemented a Mini-ELF loader and CPU/virtual-memory simulator** in C, mapping program segments into simulated memory and executing Y86 instructions.
- **Parsed binary executable headers, decoded instructions, and applied low-level memory-management and CPU-execution concepts** while debugging system-level behavior.
- **Developed the project iteratively in a sprint-based scrum-style team environment** using version control and collaborative development practices.

DNS Client/ Server — C | Systems & Network Programming

- **Built a DNS client/server implementation in C using UDP sockets**, constructing, transmitting, parsing, and responding to DNS wire-format messages.
- **Implemented binary DNS packet handling** including headers, questions, resource records, network-byte-order fields, domain-name encoding/decoding, and compressed-name pointers.
- **Loaded DNS resource records from a binary datastore into an in-memory lookup structure** and generated responses for record types including A, AAAA, CNAME, MX, NS, PTR, and SOA.

JMU Dining Web Application — Python, PostgreSQL, Jinja | Database Systems Project

- **Collaborated on a database-backed dining web application** modeled around JMU dining services, implementing backend and database functionality using Python, PostgreSQL, and Jinja.
- **Integrated relational data with server-rendered frontend views** and collaborated on application usability, query behavior, and database-backed features.

TECHNICAL SKILLS

- **Languages**: TypeScript, JavaScript, Python, C, Java, SQL, HTML/CSS
- **Frameworks & Libraries**: React, Next.js, Express.js, Node.js, Tailwind CSS, Zod
- **Data & Authentication**: PostgreSQL, Supabase Auth
- **Infrastructure & Tools**: Docker, Portainer, Cloudflare, Git, GitHub
- **Systems:** Linux/POSIX, Processes, Signals, Networking/Sockets, Virtual Memory, x86/x86-64 Assembly